

Fault Diagnosis of Gearbox Based on Improved Transformer

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ABSTRACT

Health services for the gearbox are essential to ensure the safety of industrial production. However, gearbox fault diagnosis remains a challenge due to complex responses caused by multiple gears. In recent years, there has been an increasing number of scholars who have embraced deep learning techniques for gearbox fault diagnosis. Compared with traditional deep networks, transformers exhibit outstanding pattern recognition capabilities, but their ability to process local information is lacking. Therefore, we propose an improved Transformer network that applies multi-scale perception layers and linear embedding to enhance the ability to capture local feature information, analyze fault features on multiple time scales, and retain the position information of the original signal. The validation was conducted on the Southeast University (SEU) gearbox dataset. The model's classification accuracy can reach 99.4% when detecting five different fault signals with a signal-to-noise ratio of 10. Experimental results have shown that the method has excellent diagnostic and anti-interference capabilities, which would be applied in engineering practice.

CCS CONCEPTS

• **Computing methodologies** → Machine learning; Machine learning approaches; Neural networks.

KEYWORDS

Gearbox, Fault Diagnosis, Deep Learning, Self-Attention, Transformer

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1 INTRODUCTION

Gearboxes are essential components that play a critical role in modern industrial systems. As the demand for precision, reliability, and durability in industrial manufacturing increases, gearboxes are widely used in various fields, including aerospace, machine tools, cars, and petrochemicals. The failure of gears during production can result in the destruction of the entire system, affect the accuracy of equipment, and increase the non-necessary stoppage time [1]. Therefore, fault diagnosis of gearboxes is important for ensuring the safe and stable functioning of mechanical systems, and it is crucial in preventing industrial safety accidents.

The use of vibration signals for gearbox fault diagnosis has gained widespread attention, and earlier methods relied heavily on manual feature extraction from the original signal [2]. However, with the development of typical signal processing techniques such as Fast Fourier transform, Wavelet Transform and Empirical Mode Decomposition, the process has become more refined. Although these methods can achieve some extent of fault recognition, expert knowledge still plays a dominant role. Furthermore, the detection process requires advanced computational skills and experience, making it challenging for unskilled field personnel. The emergence of artificial intelligence has led to the development of machine learning methods, which have significantly advanced fault diagnosis models [3]. Algorithms such as support vector machines, decision trees, and random forests have replaced the traditional manual feature extraction process, resulting in encouraging diagnostic results. However, traditional machine learning methods still require complex feature extraction and selection processes, that significantly impact the accuracy of diagnosis.

With the development of artificial intelligence technology, deep learning has begun to be applied in the field of gearbox fault diagnosis [4]. Applying deep learning fault diagnosis methods can reduce manual feature extraction and achieve end-to-end problem-solving,

reducing reliance on expert experience [5]. For example, Janssens et al. [6] were the first to use convolutional neural networks (CNN) for diagnosing faults in gearboxes, including bearings and gears. Their study demonstrated that deep learning algorithms can effectively extract signal features and classify them. Gong Wenfeng et al. [7] conducted experiments to analyze the influence of deep learning skills and model hyperparameters in CNN on the fault diagnosis results of rolling bearings. They proved that this method has good operability and versatility. Talebi et al. [8] proposed a dynamic modeling approach based on RNN for wind turbine generator fault diagnosis, which improves the accuracy of fault diagnosis by comparing the output results of the model and the actual system. The experiment shows that this scheme can quickly obtain fault diagnosis results and have excellent diagnostic effectiveness. Liu et al. [9] presented a method for fault diagnosis using an RNN autoencoder, which showed improved robustness in noisy environments. This method was later improved upon with the use of CNN and RNN, leading to significant performance gains. Additionally, a short and long-term memory (LSTM) network was proposed for processing one-dimensional time series data while maintaining long-term dependence. However, these technologies have certain limitations, as CNN relies on convolutions to extract features and is influenced by kernel size, which can only extract local information. RNN treats the input signal as an ensemble, limiting the parallel processing of training samples and causing problems such as gradient disappearance.

To address this issue, Vaswani et al. proposed the Transformer [10] model. Unlike traditional neural network structures, the Transformer model primarily relies on attention mechanisms to weigh and average each input sequence, establishing relationships between different sequences. It is more flexible when processing different-length sequences. Initially, Transformer is proposed to be mainly applied in the field of natural language processing (NLP), the model has since been widely adopted in the computer vision field. Specifically, the Vision Transformer (ViT) model, proposed by scholars, enabled networks to focus on different regions of an image and ultimately integrate information from the entire image. However, the Transformer model itself still has certain limitations, due to the inherent inductive bias, its ability to process local information is lacking.

An improved Transformer gearbox fault diagnosis method is proposed in this article by adding a Multi-Scale Perception layer before the Transformer layer, enriching fault information with convolutional kernels of different scales, and adding a linear mapping layer to improve network recognition accuracy. Finally, the effectiveness of the model was verified through experiments.

2 RELATED THEORIES

The basic unit of Transformer is its encoder. By using multiple encoders to stack and extract features from the raw data. This approach enables the encoder to capture the internal dependencies between sequences, thus overcoming the distance effect inherent in input sequences. Furthermore, the encoder avoids circular processing by enabling more parallelized computations on the samples, significantly improving the computational efficiency. The Transformer

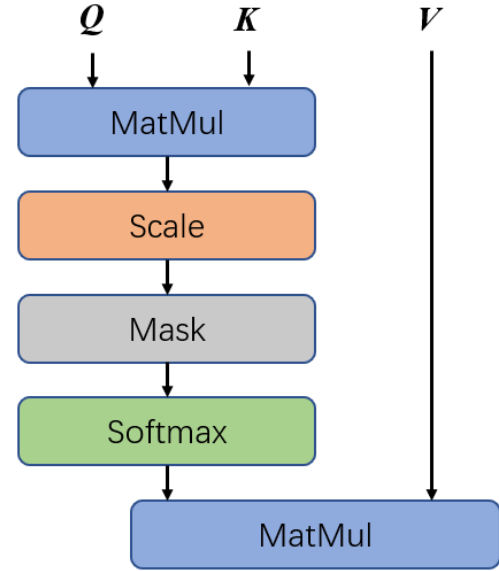


Figure 1: The basic structure of self-attention mechanism in Transformer.

model with attention mechanism has powerful parallel computing and model expression capabilities, which can effectively learn feature information and recognize patterns from samples.

The Transformer model's significant improvement is due to the introduction of the Self-attention mechanism. Figure 1 shows the basic structure of this mechanism. Through self-attention, the various information in the sequence data can be effectively associated. In particular, the self-attention mechanism performs matrix multiplications on the input vectors respectively and updates the weight matrix to obtain the correlation degree between the vectors. For vectors a_1 and a_2 , the weight matrix is multiplied to obtain the Q matrix and K matrix, and the matrix a is obtained by dot product of Q and K matrix, which represents the correlation degree between two vectors. Divide a by the scale factor k_d , the attention matrix is then processed as a normal distribution, so as to stabilize the gradient during training. Additionally, an optional mask is added, and the resulting matrix is multiplied by the weight matrix V to obtain the final output b_1 , as shown in Formula 1.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

The dimensions of Q and K are k_d , and the dimensions of V are v_d . After calculation, each feature vector in the output sequence can obtain information about all elements in the input sequence, and pay attention to important details through the assigned weights.

The purpose of adding a mask in the self-attention mechanism is to conceal certain unimportant values to prevent them from affecting the final output. For example, when padding zeros to align the sequence of samples, the positions filled with zeros do not contain important information. The specific operation of implementing the mask is to add a large negative number, and the weight coefficient calculated after Softmax is close to 0.

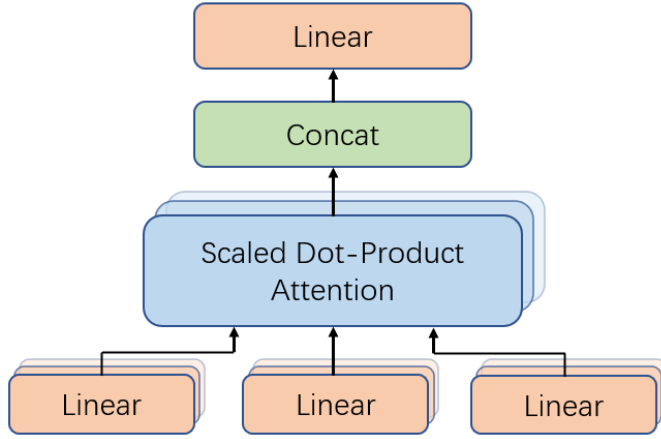


Figure 2: The basic structure of Multi-head attention block.

Unlike the basic attention mechanism that directly maps Q, K, and V to high-dimensional space, the Transformer model uses the Multi-head self-attention (MSA) to map Q, K, and V to different subspaces of high-dimensional space. MSA consists of multiple parallel self-attention layers, and the basic structure of MSA is shown in figure 2. Perform different h-times linear projections on Q, K, and V, and then calculate the self-attention function in parallel for each projection version. The output values with dimensions v_d are concatenated and reprojected to obtain the final value. The processing process can be represented as follows:

$$\begin{aligned} \text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \\ \text{head}_i &= \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \end{aligned} \quad (2)$$

In which $W_i^Q \in R^{d_{\text{model}} \times d_k}$, $W_i^K \in R^{d_{\text{model}} \times d_k}$, $W_i^V \in R^{d_{\text{model}} \times d_k}$ and $W_i^O \in R^{d_{\text{model}} \times d_k}$ can be learned in backpropagation. In general, the size of each head in MSA is $d_k = d_v = d_{\text{model}}/h$. Essentially, MSA is a combination of multiple self-attention mechanisms that discover relationships between sequences from different perspectives. This enables the model to enhance the expressive ability of each attention layer without increasing the total number of model parameters and extends the Transformer model's ability to focus on different positions.

Transformer structure is effective in acquiring global information and improving the expressive power of algorithms. However, applying Transformer solely to extract features of gearbox leads to the loss of the ability to capture local features and positional information, which prompts us to develop a better model to improve Transformer.

3 PROPOSED METHOD

In response to the shortcomings of Transformer, an improved deep learning model is proposed for gearbox fault diagnosis. The structure diagram is shown in Figure 3, where the input signal is first passed through a multi-scale perception layer, which utilizes three different-sized convolutional kernels to enhance the ability to capture local feature information. After passing through the multi-scale perception layer, the signal enters the linear embedding for linear

mapping, and a position information matrix is added to the linear embedding layer to solve the problem of Transformer losing position features. The final representation is obtained by sending the generated sequence signal to the Transformer encoder, which is then used to classify and diagnose faults.

The parameters settings for improved Transformer are shown in table 1. Here are the details of this experimental method:

3.1 Multi-scale Perception Layer

The multi-scale perception layer is designed with various convolutional kernels of different scales, which effectively improves the network's feature extraction capability. Let the input signal be represented by $X \in R^{W \times 1 \times 1}$, where W is the length of each sample. The multi-scale perception layer executes several convolution operations on the raw signal using various kernel size (101, 201, and 301, which are used in this study). The output of each convolutional kernel is considered as a channel output, and the features from each branch are added together to obtain the multi-scale features, which can be expressed as:

$$X' = [\sigma(f^{n_1 \times 1; C}(X)); \sigma(f^{n_2 \times 1; C}(X)); \sigma(f^{n_3 \times 1; C}(X))] \in R^{W \times 1 \times 3C} \quad (3)$$

Where $f^{n_i \times 1; C}(X)$ represents the size of the convolutional kernel in the convolutional process as $n_i \times 1$, and the filter number is C. $\sigma()$ indicates that ReLU activation function is used.

3.2 Linear Embedding

Divide multi-scale features into dimensions $P \times 1 \times 3C$ to form m signal sequences, where $m = W/P$, and perform linear mapping through a linear embedding layer. A self-learning matrix $E \in R^{3PC \times D}$ is used to flatten the existing feature signals into dimension D. To improve the subsequent classification work, the matrix is labeled with X_{class} , and the self-learning label is added to the flattened matrix for better subsequent classification work.

Since the position information of the sequence signal needs to be captured in the attention mechanism, the signal processed by the linear embedding layer still needs to retain the spatial information in the original signal. Add the position information matrix $E_{\text{pos}} \in R^{(m+1) \times D}$ to the signal sequence to obtain the final result z_0 of the linear embedding layer. The generated signal sequence with markers can be expressed as:

$$z_0 = [x_{\text{class}}; x_1E; \dots; x_mE] + E_{\text{pos}} \quad (4)$$

The added position information is also used as parameters in the backpropagation process for training, which effectively improves the stability of the model. Subsequently, the signal enters the Transformer encoder and passes the results of the head classifier to the classifier module to complete the final classification.

3.3 Classifier

The head classifier is comprised of a normalization layer and a fully connected layer that utilizes softmax regression to generate probability classification for each health state of the gearbox. This classification can be represented as:

$$y = FC(\text{LN}(x)) \quad (5)$$

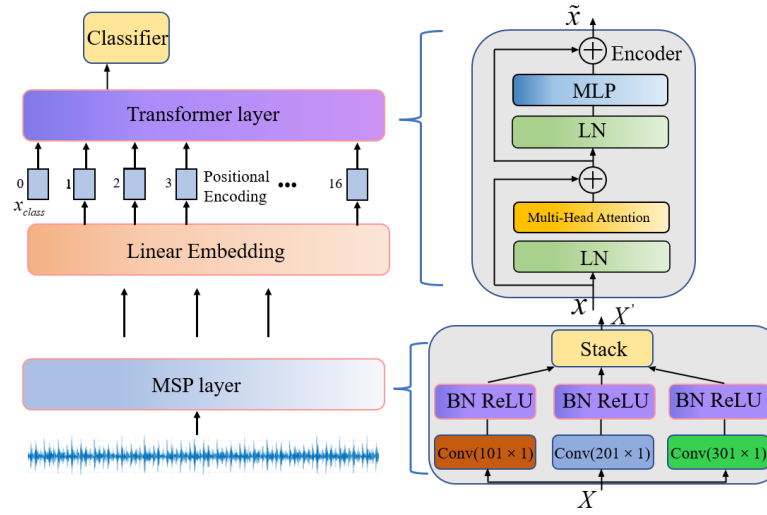


Figure 3: Improved Transformer structure diagram.

Table 1: Parameters of Improved Transformer

Number	Type	Parameters			Output
1	Input signal	-			1*1024
2	MSP layer	Kernel size	Stride	Channel	12*1024
		101/201/301	1	4	
3	Linear	Patch size	Input_dim	Output_dim	16*64*1
	Embedding	64*1	768*16	64	
4	Transformer	Depth	Head	Mlp_dim	16*64*1
	Encoder	6	16	128	
5	Classifier	Splicing dimensions			5
		64*1			

Table 2: Labeling rules for gearbox data

Category	Label	Health status description	
		Fault type	Working conditions
1	S1	Normal	20Hz-0V \ 30Hz-2V
2	S2	Chipped	20Hz-0V \ 30Hz-2V
3	S3	Miss	20Hz-0V \ 30Hz-2V
4	S4	Root	20Hz-0V \ 30Hz-2V
5	S5	Surface	20Hz-0V \ 30Hz-2V

4 EXPERIMENTS AND RESULTS

4.1 Dataset Description

The SEU gearbox dataset is provided by Southeast University [11] for the purpose of gearbox fault diagnosis. The dataset was collected using the Drivetrain Dynamics Simulator (DDS). Experimental device is shown in figure 4, which consists of brake, planet gearboxes, parallel gearboxes, and driving motors. In addition, two three-axis accelerometers (x, y, z) were installed on the parallel gearbox and

planet gearbox to collect vibration signals. The gearbox faults included Chipped, Miss, Crack in the root of gear feet, and wear in the surface of gear.

Two different working conditions are investigated with the rotating speed system load set to be either 20 HZ-0V or 30 HZ-2V. The dataset contains four fault labels and one normal label, making a total of five status data. Table 2 lists the fault data labeling rules at different speeds and loads.

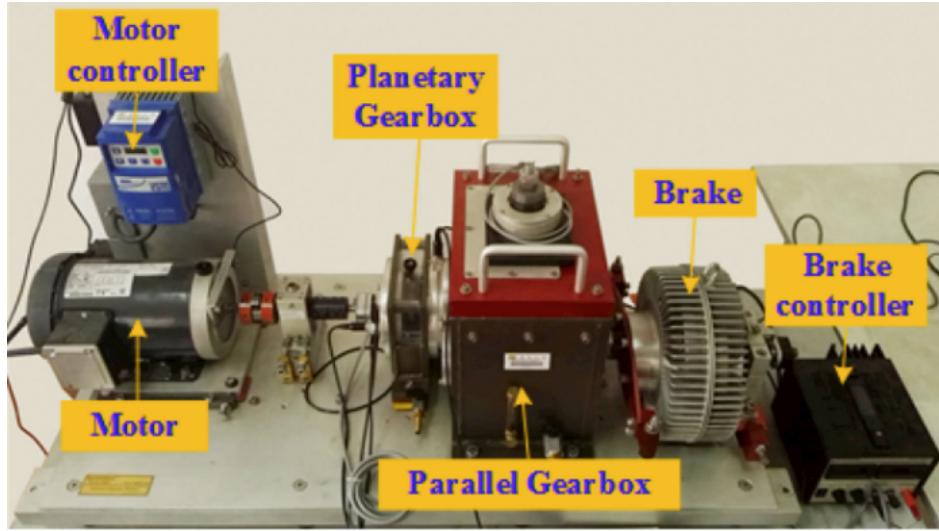


Figure 4: Experimental system for gearbox dataset.

4.2 Experiment Result and Analysis

To evaluate the fault diagnosis ability of a model on the SEU dataset, 80% of the dataset was used for training and the remaining 20% for testing. The k-fold cross-validation method was used to assist in model assessment, with k set to five to reduce random error. After approximately 50 epochs, the accuracy reached almost 99.5%. This indicates that the improved Transformer has good feature extraction ability and diagnostic accuracy.

In order to simulate the actual working conditions, noise is added to the signal to increase the difficulty of classification. Gaussian white noise based on the signal-to-noise ratio (SNR) is added. The energy relationship between signal and noise is shown in Formula (6). P_{signal} and P_{noise} represent the energy of the original signal and noise respectively. The SNR determines the amount of noise added, with SNR equal to zero indicating that the signal and noise have equal energy.

$$SNR_{dB} = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right) \quad (6)$$

Figure 5 displays a comparison of the diagnostic capabilities of the proposed model, as well as the basic models CNN and Transformer, under conditions of noise. The model exhibits a significant improvement compared to the baseline, with higher diagnostic accuracy than the original CNN and Transformer models across various noise levels. Notably, the model can achieve an impressive diagnostic accuracy of 95% when the SNR is greater than 4, demonstrating its robustness to noise. At an SNR of 10, the model's final accuracy reaches 99.4%, indicating the effectiveness and excellence of the proposed model for fault diagnosis of gearboxes.

We utilize the T-SNE algorithm to visualize the training process in two dimensions by clustering the outputs of each layer to visualize the learning process of the model. Figure 6 shows the detailed process of network training, where different colors represent different categories. It can be observed that as the network progresses, similar features continue to gather and eventually form distinct

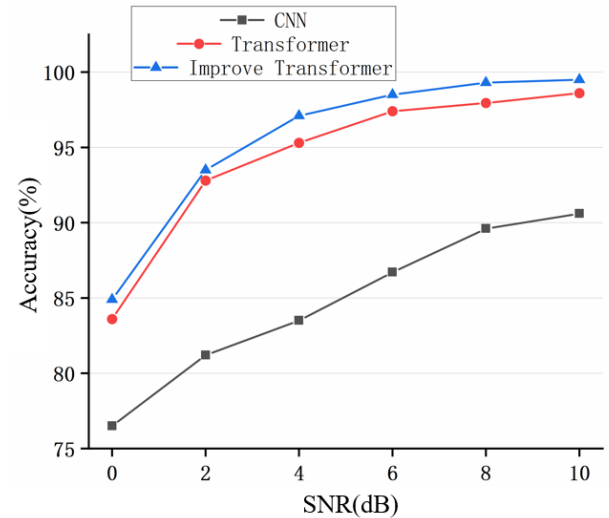


Figure 5: Diagnostic result of experimental model and baseline under different noise levels.

cluster structures. The T-SNE algorithm thus demonstrates the powerful feature learning ability of the improved model in detecting fault patterns.

Figure 7 displays the confusion matrix for each algorithm in the absence of noise. For CNN, the overall accuracy is ordinary, with some misclassifications in each class. The Transformer model exhibits low discrimination between miss teeth and wear occurring in the surface of gear, with all false positives classified as surface cracks. In contrast, the improved Transformer model exhibits the lowest error rate, indicating that the proposed model has good classification ability and has great application value in fault diagnosis.

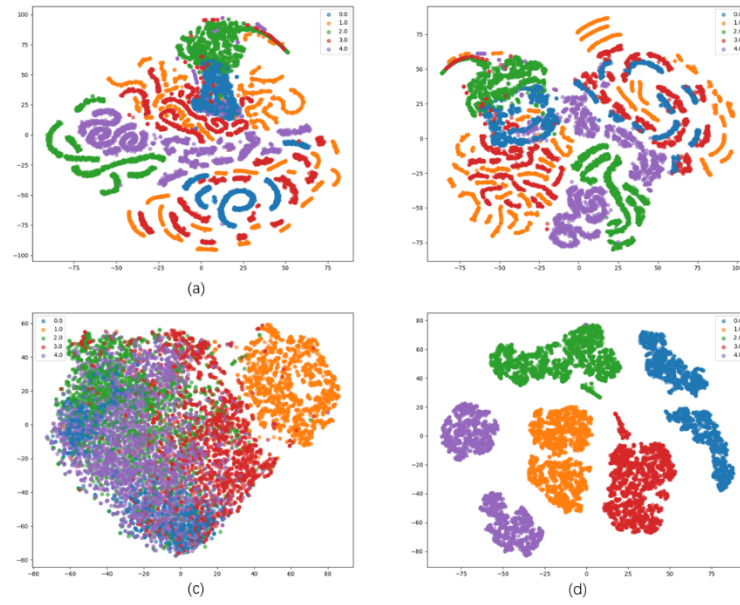


Figure 6: Display of t-SNE results for each layer of the experimental: (a) MCF, (b) Linear embed, (c) Transformer, (d) Classifier.

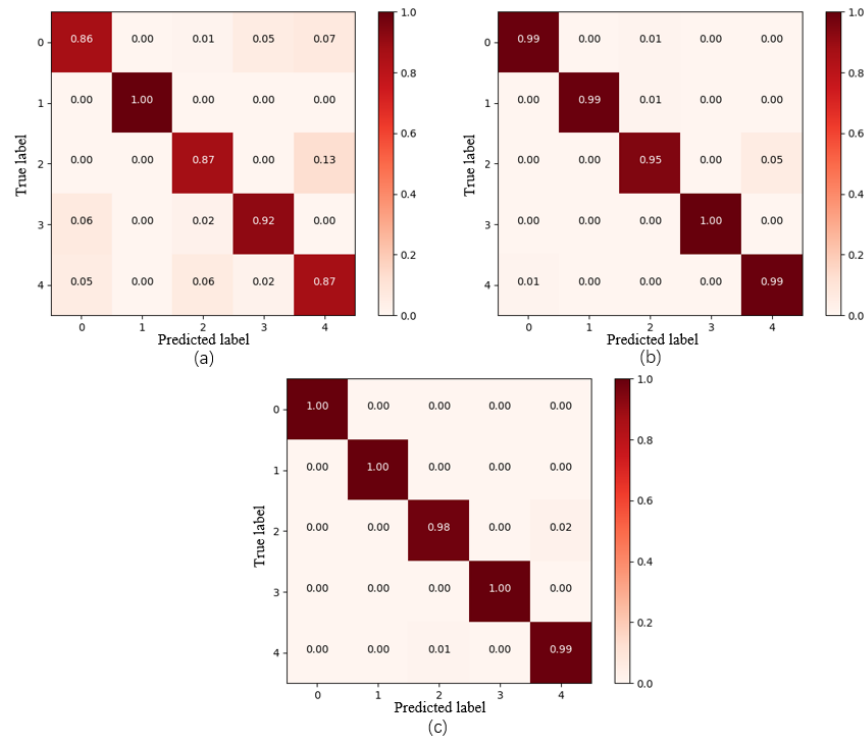


Figure 7: Confusion matrix of proposed method and baseline: (a) CNN, (b) Transformer, (c) Improved Transformer.

5 CONCLUSION

This article presents an enhanced Transformer model for intelligent fault diagnosis of gearboxes. The input signal passed through

a multi-scale perception layer, linear embedding and Transformer encoder to finally get the classification result. The SEU gearbox dataset was utilized for testing, and the results of the experiments demonstrate that the improved Transformer model is capable of

effectively capturing local feature information through the integration of multi-scale perception layers and linear embedding layers. The proposed model has the ability to analyze fault features at various time scales while retaining the position information of the original signal. During the experiment, the model outperformed other basic models in terms of diagnostic capabilities. The model's classification accuracy can reach 99.4% when detecting five different fault signals with a signal-to-noise ratio of 10. Even with a signal-to-noise ratio of 2, the model can still maintain a diagnostic accuracy of 95%. These results demonstrate the model's exceptional anti-interference ability, suggesting that the proposed method could be effective in real-world engineering fault diagnosis tasks.

In the future, we plan to use more advanced models to improve the proposed method. For example, using the recent RetNet, which has faster speed and accuracy than Transformer. We will also collect more data for different gearbox faults to evaluate the performance of the proposed method.

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